Work through the problems below. Draw pictures where they help — a quick Venn diagram often makes an "and/or/not" question obvious.

Part 1: Set Theory — Unions, Intersections, Complements

1. Roll one fair six-sided die, so $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let A be the event the roll is **even** and B be the event the roll is **4 or greater**.

   (a) List the outcomes in A and in B.
   (b) List the outcomes in $A \cup B$.
   (c) List the outcomes in $A^C \cap B$.
   (d) List the outcomes in $(A \cup B)^C$.

2. Draw one card from a standard 52-card deck. Let H be the event the card is a **heart** and F be the event the card is a **face card** (Jack, Queen, or King). Write each event below in set notation using H, F, $\cap$, $\cup$, and ^{C}.

   (a) The card is a heart **and** a face card.
   (b) The card is a heart **but not** a face card.
   (c) The card is **neither** a heart **nor** a face card.
   (d) Shade in a Venn diagram to describe the region in part (c).

3. Draw a Venn diagram below for two events A and B inside an outcome space Ω. On separate copies of the diagram:

(a) Shade the region $A \cap B^C$ (in A but not in B).
(b) Shade the region $(A \cup B)^C$.
(c) Shade the outcomes that are in **exactly one** of A or B, and then write that region in set notation.

4. Consider an outcome space Ω with events A, B that are *not* mutually exclusive. Let $P(A) = 0.5$, $P(B) = 0.7$, and $P(A \cap B) = 0.4$.

   (a) Draw a Venn diagram with two circles, one for A and one for B.
   (b) Using your diagram, calculate $P(A \cup B)$. Explain why you cannot simply add $P(A)$ and $P(B)$ to get this answer.
   (c) Find $P\left(A^C\right)$.
   (d) Find $P\left(A \cap B^C\right)$, the probability of "A but not B."

5. For each pair of events below, state whether the two events are **mutually exclusive** (recall this means $A \cap B = \emptyset$).

   (a) Draw one card: "the card is a heart" and "the card is a spade."
   (b) Draw one card: "the card is a heart" and "the card is a king."
   (c) Roll one die: "the roll is even" and "the roll is odd."
   (d) Roll one die: "the roll is even" and "the roll is 4 or greater."

Part 2: The Axioms of Probability

6. **Inclusion–exclusion.** In a survey, the probability a person drinks coffee is 0.7, the probability they drink tea is 0.4, and the probability they drink **both** is 0.2.

   (a) Define events A = "person drinks coffee" and B = "person drinks tea." State whether the problem gives you $P(A)$, $P(B)$, $P(A \cap B)$, and $P(A \cup B)$, and write down each value it does give.
   (b) Find the probability a person drinks coffee **or** tea.
   (c) Find the probability a person drinks **neither**.
   (d) Find the probability a person drinks coffee **but not** tea.

7. **Which axiom breaks?** For each proposed assignment of probabilities, decide whether it is *possible*. If it is impossible, name the axiom or consequence it violates and explain.
   (a) $P(A) = 0.6$, $P(B) = 0.7$, and A and B are mutually exclusive.
   (b) $P(A) = -0.2$.
   (c) $P(A) = 0.6$, $P(B) = 0.7$, and $P(A \cap B) = 0.4$.

(BREAK, END OF FIRST PART OF CLASS)

Part 3: Conditional Probability

8. The table below gives the **joint probabilities** for a randomly chosen commuter: their main mode of transport (Bike, Bus, or Car) and whether they arrived **Late** or **On time**. Each entry is the probability of that combination; the margins give the totals.

	Bike	Bus	Car	**Total**
Late	0.05	0.15	0.05	**0.25**
On time	0.20	0.25	0.30	**0.75**
Total	**0.25**	**0.40**	**0.35**	**1.00**

   (a) Find $P(\text{Late})$ and $P(\text{Bus})$.
   (b) Find $P(\text{Late} \cap \text{Bus})$.
   (c) Find $P(\text{Late} \mid \text{Bus})$.
   (d) Find $P(\text{Bike} \mid \text{Late})$ and $P(\text{Bike} \mid \text{On time})$. Is a commuter more likely to bike when they are late, or when they are on time?
   (e) Is the event "Late" independent of the event "Bus"? Compare $P(\text{Late} \mid \text{Bus})$ to $P(\text{Late})$ to decide.

9. Two events A and B satisfy $P(A) = 0.3$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$.
   (a) Find $P(A \mid B)$. Did conditioning on B make A **more** or **less** likely than its original probability $P(A)$?
   (b) Find $P\left(A \mid B^C\right)$. (Hint: first find $P\left(A \cap B^C\right)$ and $P\left(B^C\right)$.)
   (c) Are A and B independent? Explain in a sentence what these numbers say about how A and B are related.

10. Tomorrow there is a 40% chance of rain. If it rains, there is an 80% chance you carry an umbrella; if it does not rain, there is only a 10% chance you carry an umbrella.
    (a) Find $P(\text{rain and umbrella})$ using the rule $P(A \cap B) = P(B)\, P(A \mid B)$.
    (b) Find $P(\text{no rain and umbrella})$.
    (c) Use parts (a) and (b) to find $P(\text{umbrella})$. (Hint: "rain" and "no rain" are mutually exclusive, so add the two pieces.)